higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

4 August 2025

This marking guideline consists of 8 pages.

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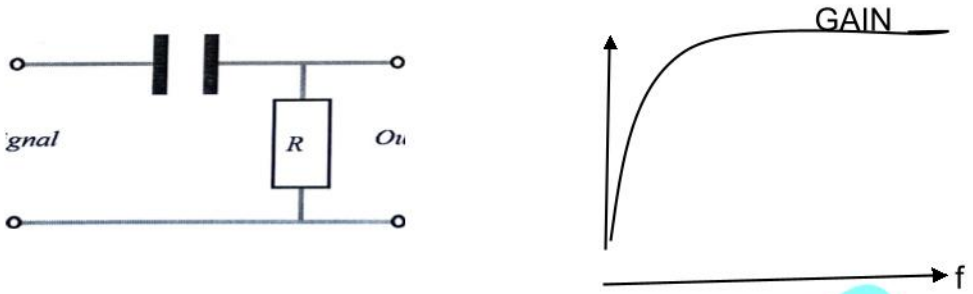
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NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	**N4** Mechanical Draughting NATED
N1-N3 Electrical Trade Theory NATED	**N2-N3** Diesel Trade Theory NATED	**N5-N6** Strength of Materials & Structures NATED	**N3** Building & Civil Technology NATED	**N1-N2** Fitting and Machining Theory NATED	**N1-N4** Plating and Structural Steel Drawing NATED
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N5-N6 Fluid Mechanics NATED	**N4-N6** Supervisory Management NATED	**N4-N6** Digital Electronics NATED	**N6** Control Systems NATED	**N1-N2** Bricklaying &Plastering Theory NATED	More…

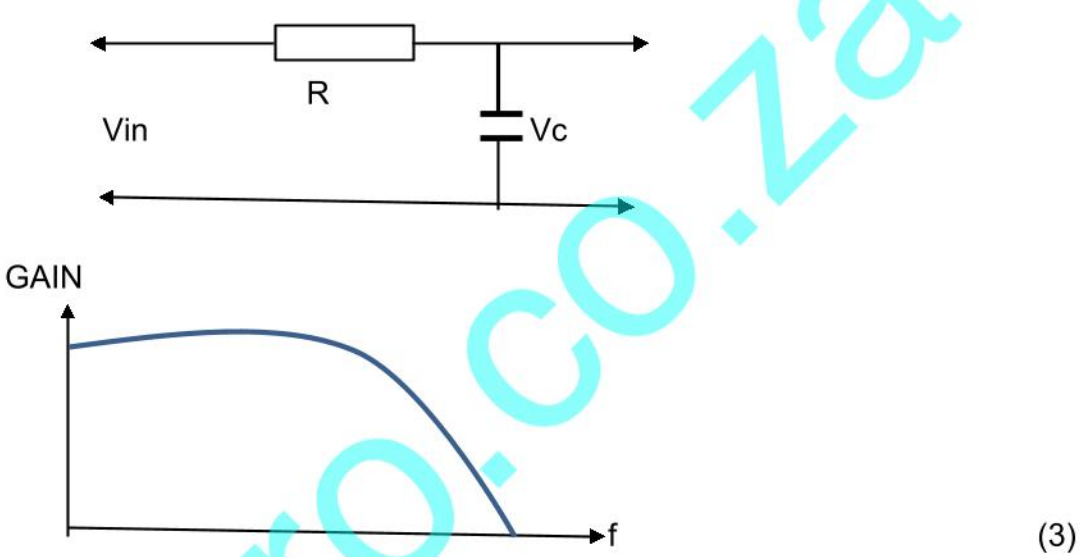
QUESTION 1: AC THEORY

1.1 1.1.1



(3)

1.1.2



(3)

1.2 1.2.1

$$I_{Z1} = \frac{V_T}{Z_1}$$

$$= \frac{25 \angle 0°}{15{,}81 \angle 71°}$$

$$= 1{,}582 \angle -71° \ A$$

$$I_{Z2} = \frac{V_T}{Z_2}$$

$$= \frac{25 \angle 0°}{23.26 \angle -64\ 54°}$$

$$= 1{,}075 \angle 64{,}54 \ A$$

$$I_T = I_{Z1} + I_{Z2}$$

$$= (0{,}515 - j\ 1{,}496) + (0{,}46 + j\ 0{,}97)$$

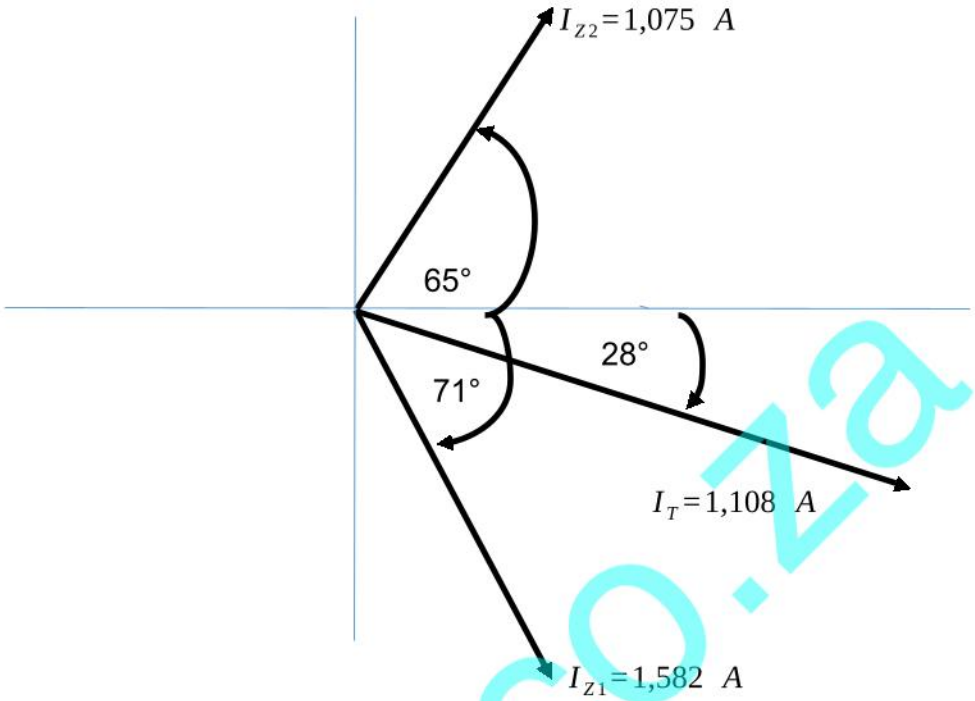
$$= 0{,}975 - j\ 0{,}526$$

$$= 1{,}108 \angle -28{,}3 \ A$$

(6)

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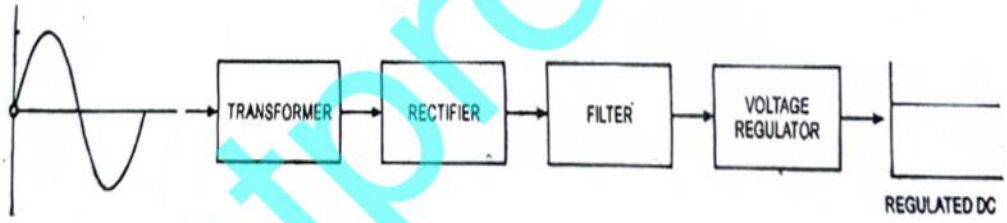
1.2.2



(3)
[15]

QUESTION 2: POWER SUPPLY

2.1



(ALLOCATE HALF MARK FOR EACH CORRECTLY LABELLED BLOCK INCLUDING INPUT AND OUTPUT WAVE)

(3)

2.2 2.2.1

$$V_{DC} = V_M - \frac{I_{DC}}{4fC}$$

$$V_{DC} = \frac{V_M}{\left(1 + \frac{1}{4fC\,R_L}\right)}$$

$$= \frac{33{,}941}{\left(1 + \frac{1}{4 \times 50 \times 2200\,x\,10^{-6} \times 220}\right)}$$

(4)

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$= 33,594 \ V$ ✓

2.2.2 $I_{DC} = \frac{V_{DC}}{R_{DC}}$

$= \frac{33,594}{220}$ ✓

$= 0.153 \ A$ ✓ (2)

2.2.3 $r = \frac{1}{4\sqrt{3} f C R_L}$

$= \frac{1}{4\sqrt{3} \times 50 \times 2200 \, x \, 10^{-6} \times 220}$

$= 0,006$ (2)

2.3 During positive half cycle across the transformer, D_1 conducts and D_2 is off. C_1 charges to Vm. During negative half cycle, D_2 will conduct and D_1 is off. C_2 charges to Vm. The voltage drop across both capacitors will therefore be 2Vm. (4)

[15]

QUESTION 3: AMPLIFIER

3.1 3.1.1 $A_i = h_{fe} = 80$ (1)

3.1.2 $A_V = \frac{-h_{fe} z_L}{h_{ie}}$

$= \frac{-80 \times 2400}{13000}$

$= 14,76$ ✓ (2)

3.1.3 $Z_1 = h_{ie} = 13 \, k\Omega$ ✓

$Z_{IN} = R_B // Z_1$

$= \frac{300000 \times 13000}{300000 + 13000}$

$= 12,46 \, k\Omega$ (2)

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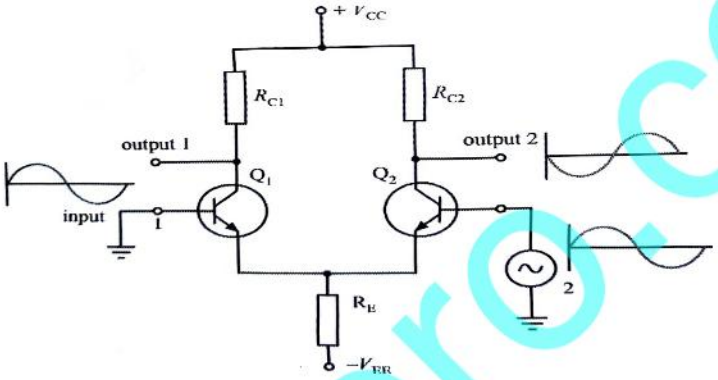
3.1.4 $$Z_2 = \frac{1}{h_{oe}} = \frac{1}{35 \times 10^{-6}} = 28{,}571 \ k\Omega$$ ✓

$$Z_{out} = R_C // Z_2$$

$$= \frac{2400 \times 28571}{2400 + 28571}$$

$$= 2{,}214 \ k\Omega$$ ✓ (2)

3.2



(Allocate one full mark for correctly labelled electrical components) (5)

3.3 $$\frac{V_O}{V_{IN}} = 1 + \frac{R_f}{R_{in}}$$

$$R_f = \frac{120 \, x \, 10^{-3}}{20 \, x \, 10^{-3}} - 1$$

$$= 10 \ k\Omega$$ (3)

[15]

QUESTION 4: IC AND TRANSDUCERS

4.1
- Very small
- Less weight
- Reduce cost
- High reliability
- Low power requirements
- Easy replacement

(Any 3 × 1)) (3)

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4.2
- Troubleshooting electronic equipment or handling printed circuit boards without using an electronic wrist strap.
- Placement of synthetic materials on or near electronic equipment.
- Rapid movement of air near electronic equipment. (3)

4.3
4.3.1 False
4.3.2 True
4.3.3 True
4.3.4 False
4.3.5 True
(5 × 1) (5)

4.4
$$R_t = A\,e^{\frac{B}{T}}$$

$$10000 = A\,e^{\frac{3600}{298}}$$

$$A = 56,6878\,x\,10^{-3}\checkmark$$

$$R_{t@30^\circ C} = A\,e^{\frac{B}{T}}$$

$$= 56,6878\,x\,10^{-3}\,e^{\frac{3600}{303}}$$

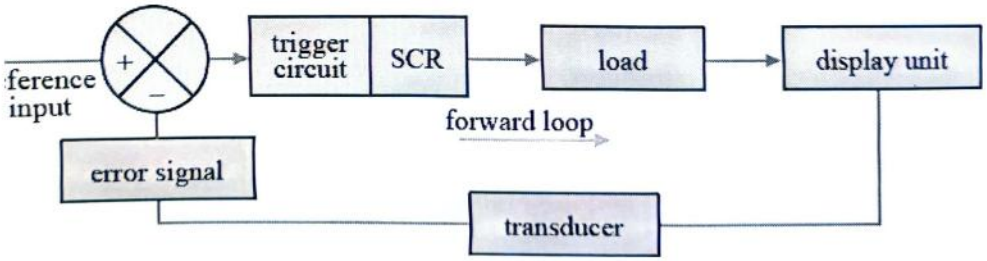
$$= 8,193\ \ k\Omega$$

(4)
[15]

QUESTION 5: ELECTRONIC PHASE CONTROL

5.1
- SCR is a semiconductor diode that will only conduct once the gate has been triggered by a pulse to set it into conduction in one direction.✓✓
- TRIAC is component that will allow conduction in both directions, both half cycles provided that the gate has been triggered by a pulse to set it into conduction.✓✓ (2 × 2) (4)

5.2



feedback loop

(4)

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(Allocate one mark for comparator, and half mark for each correctly labelled block in the diagram)

5.3
- Tacho generator
- Synchronous motor (2)

[10]

QUESTION 6: MEASURING INSTRUMENTS

6.1 Digital signal is a signal that will vary between two definite values as time progresses.✓✓ (1 × 2) (2)

6.2
- The input voltage is attenuated and then applied to a null detector.
- When the measurement starts, the logic control circuit sends a reset signal, which resets the counter and the DAC.
- The output of the counter is fed to the DAC.
- With each count, the ramp generated by DAC increases incrementally, producing the staircase ramp as the output.
- The null detector receives the input voltage and the staircase ramp voltage produced by DAC as input.
- The ramp is continuously compared with the input signal.
- The ramp increment continues until it matches the input voltage.
- Once the two voltages are equal, the null detector produces a signal, triggering the logical control circuit. (6 × 1) (6)

6.3
- Continuous balance digital voltmeter
- Double slope voltmeter
- Successive approximation voltmeter
- Stair slope voltmeter (Any 2 × 1)) (2)

[10]

QUESTION 7: OSCILLATORS

7.1
- Change in load conditions
- An unstable supply voltage
- Variations in the tuned circuit and transistors parameters. (3)

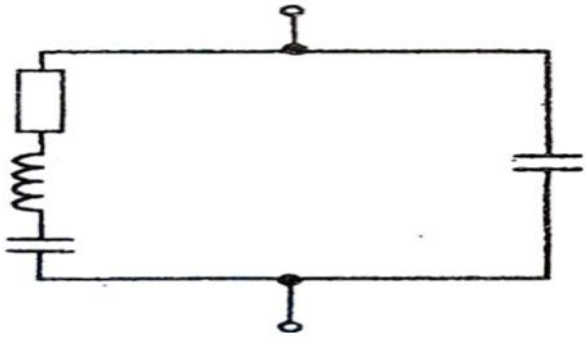
7.2
- When Vin is close to zero, Q_1 is OFF while Q_2 is ON, placing V_{out} at the mid-point of the potential divider.
- As V_{in} decreases, Q_1 starts to turn ON, starving Q_2 of base current and causing it to turn OFF.
- This process is hastened by positive feedback, leading Q_1 to saturate and Q_2 to turn OFF completely.
- As v_{in} decreases back towards ZERO, Q_1 begins to turn OFF, allowing Q_2 to turn ON again gradually. (3)

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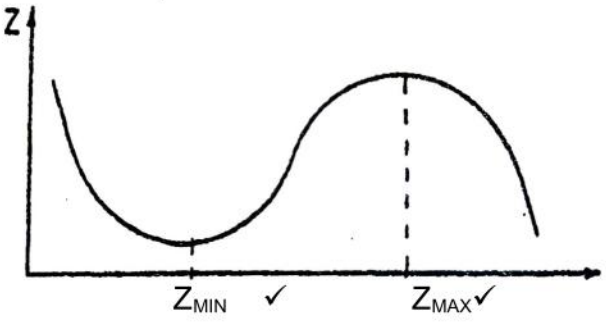
- Positive feedback aids this transition, causing Q_1 to turn OFF completely and Q_2 to saturate, thus shifting V_{out} back to its initial state.

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7.3



Allocate 1 full for series and and 1 mark for parallel circuit)



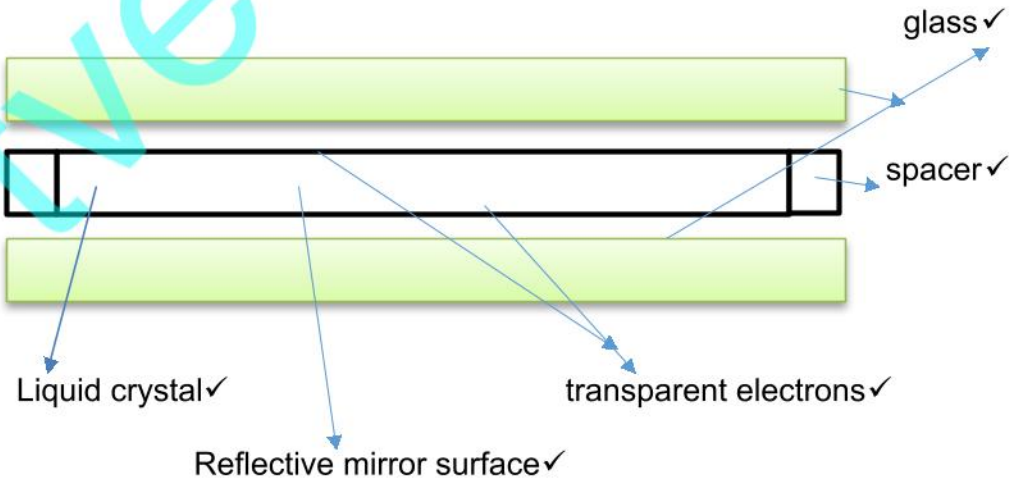
(4)
[10]

QUESTION 8: LIQUID CRYSTAL DISPLAY

8.1
- Consume less power
- Cost low
- Provide excellent contrast
- Thin and lighter weight
- Long lifespan
- No burnt-in images

(Any 5 × 1) (5)

8.2



(5)
[10]

TOTAL: 100

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